

1. Data Management

1.1 Experimental Protocol

The dataset contains approximately 3,500 lung X-ray images distributed across three classes: Normal (1,250), Opacity (1,125), and Pneumonia (1,100). The distribution is mildly imbalanced, which informed the choice of experimental protocol.

Stratified 5-Fold Cross-Validation (StratifiedKFold, $k=5$, $\text{random_state}=42$) was adopted. Stratified splitting ensures each fold maintains the original class proportions, which is important given the class imbalance. The first fold was used for training (80%, ~2,800 images) and validation (20%, ~700 images). This protocol maximises data utilisation while providing a reliable estimate of generalisation performance.

1.2 Pre-processing

All images were resized from 299×299 to 224×224 pixels to match the standard input size of ResNet-18. Pixel values were normalised using ImageNet statistics ($\text{mean}=[0.485, 0.456, 0.406]$, $\text{std}=[0.229, 0.224, 0.225]$). Although X-ray images differ from natural photographs, applying ImageNet normalisation is appropriate when using a pretrained ResNet backbone, as it aligns the input distribution with what the network originally learned from.

1.3 Data Augmentation

Augmentation was applied to the training set only, not the validation set. The following transforms were applied:

- Random Horizontal Flip ($p=0.5$): Lung anatomy is approximately bilaterally symmetric, so flipping produces anatomically plausible variants.
- Random Rotation ($\pm 5^\circ$): Simulates minor variation in patient positioning during image acquisition.

More aggressive transforms such as vertical flipping or large shear were deliberately avoided as they would produce anatomically implausible images that could mislead the classifier.

2. Neural Networks

2.1 Proposed Custom CNN

The CustomCNN is a bespoke convolutional architecture designed to balance classification performance with parameter efficiency. It comprises four convolutional blocks with progressively increasing channel depth (32 → 32 → 64 → 64 → 128 → 128 → 256). Batch Normalisation follows the first convolution in each block to stabilise gradient flow, while Max Pooling halves the spatial resolution after every two layers. A final AdaptiveAvgPool2d(1,1) collapses the spatial dimensions entirely, replacing large fully-connected layers with a compact 256-dimensional feature vector. The classifier consists of two linear layers separated by Dropout (0.5 and 0.3) to regularise against overfitting. The total parameter count is approximately 340K, considerably fewer than any standard architecture.

2.2 Existing Architecture: ResNet-18

ResNet-18 was selected as the existing architecture due to its favourable balance of depth, representational capacity (~11.18M parameters), and well-established performance on medical imaging tasks. Its residual (skip) connections mitigate the vanishing gradient problem and make it well-suited to transfer learning. The final fully-connected layer was replaced with a new linear layer mapping from 512 features to 3 output classes across all strategies.

Three training strategies were investigated:

- Trained from Scratch: All weights randomly initialised; no pretrained knowledge. Serves as a baseline to assess whether ImageNet pretraining provides benefit.
- Fine-Tuning: ImageNet pretrained weights loaded; all layers remain trainable. The entire network adapts to the lung X-ray domain, leveraging low-level features such as edges and textures already encoded by pretraining.
- Feature Extractor: ImageNet pretrained weights loaded; backbone frozen (requires_grad=False). Only the new classification head is trained. This is the most parameter-efficient transfer strategy, training only 1,538 parameters.

2.3 Training Components

Loss function: Cross-Entropy Loss was used for all models, as it is the standard choice for multi-class classification and directly optimises the log-likelihood of the correct class.

Optimiser: Adam (lr=0.001) was chosen for all models. Adam's adaptive per-parameter learning rates enable faster convergence than SGD on smaller datasets, and it is less sensitive to learning rate tuning, making it appropriate here.

Epochs: The CustomCNN was trained for 20 epochs and each ResNet strategy for 20 epochs, providing a consistent and fair comparison across all models.

2.4 Results Summary

Table 1 reports validation accuracy, macro F1-score, and trainable parameter count for all models. Values shown are the best validation accuracy achieved by each model during training; Macro F1, Cohen's Kappa, ROC-AUC and MCC were computed on the held-out test set for the selected best model (ResNet-18 Fine-Tuning).

Model	Strategy	Val Acc	Macro F1	Params
CustomCNN	Scratch	0.894	n/a*	~340K
ResNet-18	Scratch	0.892	n/a*	11.18M
ResNet-18	Fine-Tuning	0.937	0.934	11.18M
ResNet-18	Feat. Extractor	0.892	n/a*	1,538

Table 1: Validation performance summary across all models and training strategies.

**Macro F1 was only computed on the held-out test set for the selected best model (ResNet-18 Fine-Tuning); per-epoch validation tracked accuracy only for the other configurations.*

ResNet-18 Fine-Tuning achieves the highest validation accuracy and macro F1, confirming that pretrained ImageNet representations transfer effectively to the lung X-ray domain. The CustomCNN, despite having ~33× fewer parameters than ResNet-18, achieves competitive performance, demonstrating that a well-designed lightweight architecture can be effective on this task. The Feature Extractor underperforms fine-tuning because the frozen backbone cannot adapt its representations to the distributional shift between natural images and medical X-rays. ResNet-18 trained from scratch outperforms the CustomCNN due to greater model capacity, but requires more epochs to converge. Based on these results, ResNet-18 Fine-Tuning is selected as the best model for final evaluation.